Lab 1: Complete Lab-Focused Teaching Package

🎯 LAB COMPLETION CRITERIA

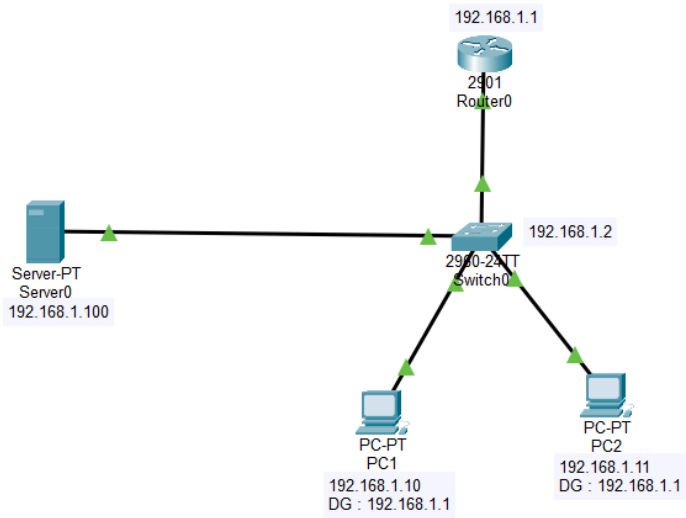
- ☑ All devices respond to ping from at least one other device
- ☑ ARP tables show correct IP-MAC mappings
- ☑ Switch MAC table shows both PCs' MAC addresses
- ☑ Simulation clearly shows encapsulation process
- ☑ All troubleshooting scenarios resolved
- ☑ Worksheets completed with accurate data

📁 PACKET TRACER FILE REQUIREMENTS

File Name: ITN_Module3_ProtocolLab.pkt Required Components:

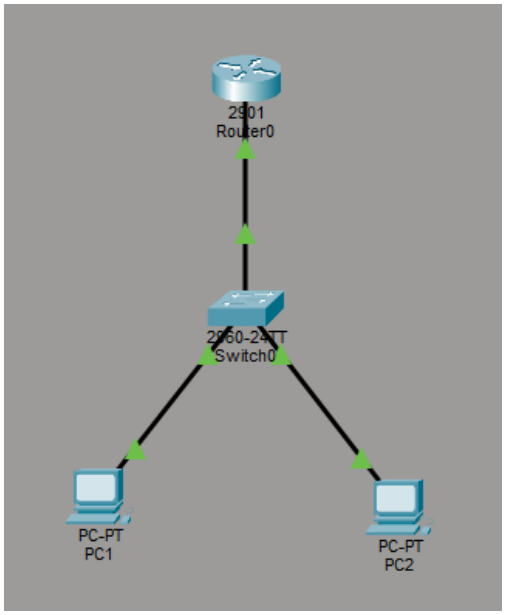
- 1× Router 2911 (R1)
- 1× Switch 2960 (S1)
- 2× PCs (PC1, PC2)
- Copper straight-through cables
- All configurations from this lab script

✅**Bonus:** Add a server at 192.168.1.100 for HTTP protocol analysis



Source: https://drive.google.com/file/d/1dewo2MTuxE8BDB0hset3nrmDEnYeyv3A/view?usp=sharing

Part 1 : Configure basic device addressing (R1, S1, PC1, PC2)



Addressing Table

Device	Interface	IP Address	Subnet Mask	Default Gateway
R1	G0/1	192.168.1.1	255.255.255.0	N/A
S1	VLAN 1	192.168.1.2	255.255.255.0	192.168.1.1
PC1	NIC	192.168.1.10	255.255.255.0	192.168.1.1
PC2	NIC	192.168.1.11	255.255.255.0	192.168.1.1

Part 2 : Verify connectivity and observe encapsulation

PC1

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0004.9aec.de02        dynamic
  192.168.1.2           0060.3e86.7626        dynamic
  192.168.1.11          0007.ec78.42cc        dynamic
```

PC2

```
C:\>ping 192.168.1.10

Pinging 192.168.1.10 with 32 bytes of data:

Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128
Reply from 192.168.1.10: bytes=32 time<1ms TTL=128

Ping statistics for 192.168.1.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms
```

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0004.9aec.de02        dynamic
  192.168.1.2           0060.3e86.7626        dynamic
  192.168.1.10          00e0.b003.a6c1        dynamic
```

R1

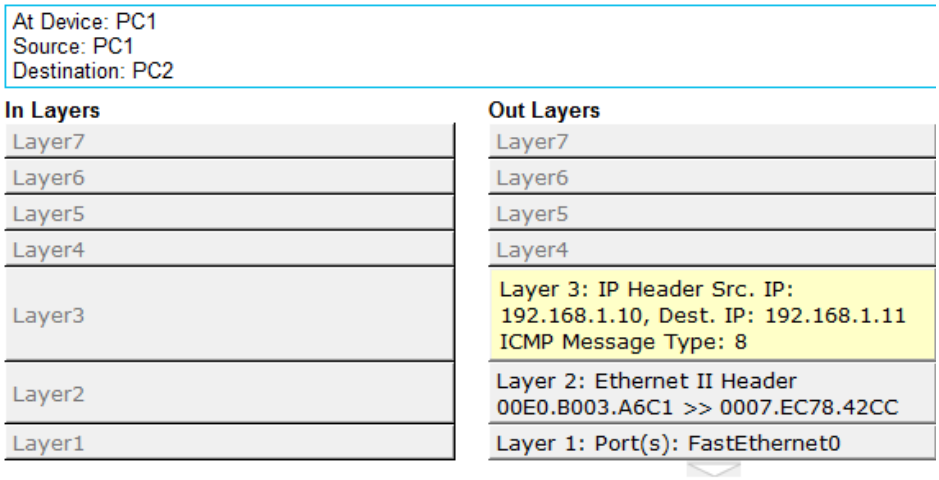
```
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES manual down                  down
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
Vlan1                  unassigned      YES NVRAM  administratively down down
```

```
R1#show arp
Protocol  Address          Age (min)  Hardware Addr   Type   Interface
Internet  192.168.1.1            -   0004.9AEC.DE02  ARPA   GigabitEthernet0/1
Internet  192.168.1.10          11   00E0.B003.A6C1  ARPA   GigabitEthernet0/1
Internet  192.168.1.11           6   0007.EC78.42CC  ARPA   GigabitEthernet0/1
```

S1

```
S1#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

Part 3 : Analyze protocol headers using simulation



คำอธิบาย : PC1 ส่ง ICMP Echo Request ไปยัง PC2 โดยมี IP ต้นทาง 192.168.1.10 และ IP ปลายทาง 192.168.1.11

ก่อนส่ง ICMP จะมีการใช้ ARP เพื่อค้นหา MAC Address ของ PC2

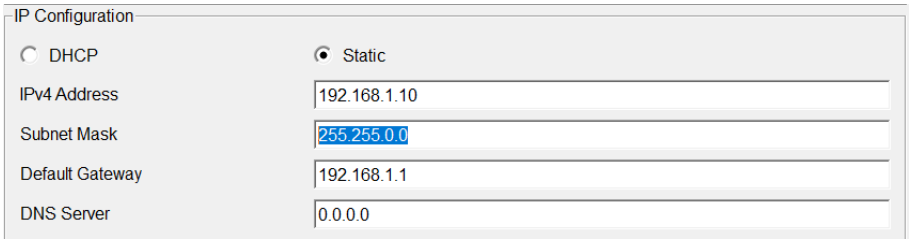
เมื่อทราบ MAC แล้ว จึงห่อหุ้มข้อมูลใน Ethernet Frame และส่งผ่านสวิตช์ S1 ไปยัง PC2

Step	Protocol	OSI Layer	TCP/IP Layer	Key Field Observed
1	ARP Request	Layer 2	Network Access	Target IP: 192.168.1.11
2	ARP Reply	Layer 2	Network Access	Sender MAC: 0007.EC78.42CC
3	ICMP Echo	Layer 3	Internet	Type: 8, Code: 0
4	Ethernet	Layer 2	Network Access	Src MAC: 00E0.B003.A6C1 ⟶ Dest MAC: 0007.EC78.42CC
5	ICMP Reply	Layer 3	Internet	Type: 0, Code: 0

Part 4 : Troubleshoot layered connectivity issues

Scenario A: Incorrect Subnet Mask

1) ตั้งค่า Subnet Mask ใหม่ที่ PC1




```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::2E0:B0FF:FE03:A6C1
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.10
   Subnet Mask.....................: 255.255.0.0
   Default Gateway.................: ::
                                     192.168.1.1

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

2) Ping ไปที่ PC2

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=1ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms
```

3) ใช้คำสั่ง arp -a

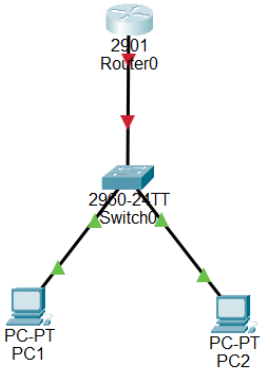
```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.11          0007.ec78.42cc        dynamic
```

จะเห็นได้ว่า ถึงแม้ Subnet mask จะไม่ถูกต้อง การแก้ไขด้วย arp ยังสามารถทำได้สำเร็จเพราะ host ทั้งสองถือว่าอยู่ในเครือข่ายเดียวกัน

Scenario B: Router Interface Shutdown

1) ปิดพอร์ตที่ router (ในไฟล์นี้ใช้ router ที่ g0/1)

```
R1>enable
R1#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES manual down                  down
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
R1#configure terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/1
R1(config-if)#shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to administratively down

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
down
```



2) ทดสอบ ping จาก PC1

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
```

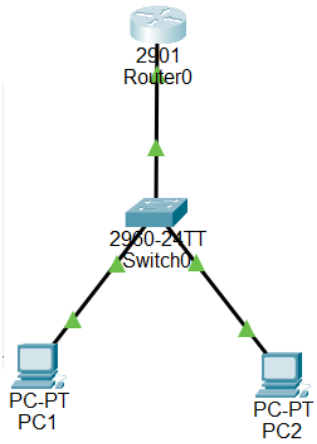
3) ตรวจสอบด้วย ARP

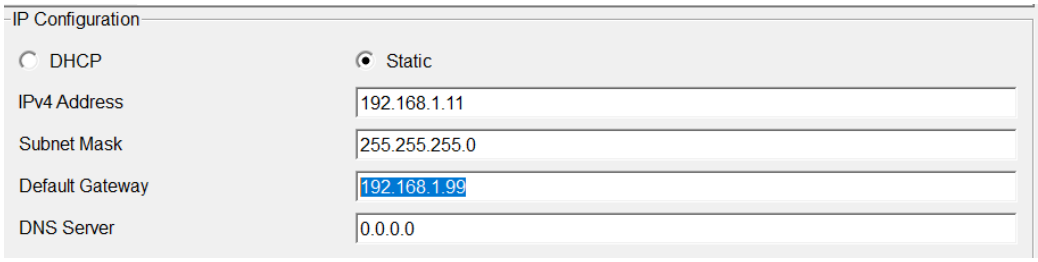
```
C:\>arp -a
No ARP Entries Found
```

Router interface ที่เชื่อมกับ LAN ถูกปิด ทำให้การเชื่อมต่อไปที่ default gateway ขาดไป

4) เปิดพอร์ตที่ router

```
R1>enable
R1#config terminal
Enter configuration commands, one per line.  End with CNTL/Z.
R1(config)#interface g0/1
R1(config-if)#no shutdown

R1(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to up
```






```
C:\>ipconfig

FastEthernet0 Connection:(default port)

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: FE80::207:ECFF:FE78:42CC
   IPv6 Address....................: ::
   IPv4 Address....................: 192.168.1.11
   Subnet Mask.....................: 255.255.255.0
   Default Gateway.................: ::
                                     192.168.1.99

Bluetooth Connection:

   Connection-specific DNS Suffix..:
   Link-local IPv6 Address.........: ::
   IPv6 Address....................: ::
   IPv4 Address....................: 0.0.0.0
   Subnet Mask.....................: 0.0.0.0
   Default Gateway.................: ::
                                     0.0.0.0
```

2) Ping จาก PC2

```
C:\>ping 192.168.1.1

Pinging 192.168.1.1 with 32 bytes of data:

Reply from 192.168.1.1: bytes=32 time=12ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time<1ms TTL=255
Reply from 192.168.1.1: bytes=32 time=1ms TTL=255

Ping statistics for 192.168.1.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 12ms, Average = 3ms

C:\>ipconfic
Invalid Command.
```

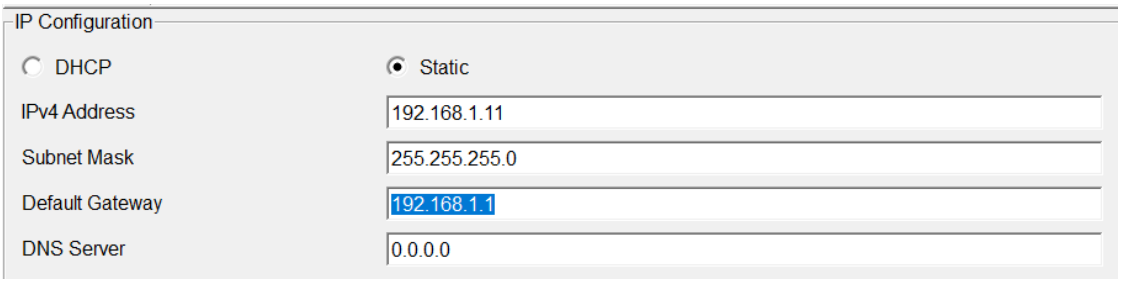
สามารถ ping ไปที่ router ได้สำเร็จเพราะปลายทางอยู่ Subnet เดียวกัน default gateway ไม่จำเป็นสำหรับ network ภายใน การเชื่อมต่อจะล้มเหลวต่อเมื่อมีการเชื่อต่อกับ network ภายนอก

3) ใช้ rount print (ในที่นี้จะใช้ arp -a แทนเพราะใช้ใน Packet Tracer)

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0004.9aec.de02        dynamic
```

ไม่มี MAC ของ 192.168.1.99

4) ตั้งค่ากลับให้เป็นเหมือนเดิม



Part 5 : Document protocol interactions

(PC1 ⟶ PC2 : ICMP Ping)

1. Application Layer

   PC1 Application

   - User executes: ping 192.168.1.11
   - Generates ICMP Echo Request data

2. Transport Layer

   (No TCP/UDP used)

   - ICMP does not use TCP or UDP

3. Network Layer

   IP Packet Created

   - Source IP: 192.168.1.10
   - Destination IP: 192.168.1.11
   - Protocol: ICMP (1)
   - TTL: 128

4. Data Link Layer

   Ethernet Frame Encapsulation

   - Source MAC: 00E0.B003.A6C1
   - Destination MAC: 0007.EC78.42CC
   - FCS added

5. Physical Layer

   Bits transmitted as electrical signals over Ethernet cable via Switch S1

PC2 (Decapsulation)

1. Physical Layer

   Bits received

2. Data Link Layer

   Ethernet frame checked (FCS)

3. Network Layer

   IP header removed

   ICMP Echo Request processed

4. Application Layer

   PC2 generates ICMP Echo Reply

   (Type 0, Code 0)

📝 LAB DELIVERABLES

Worksheet 1: Configuration Verification

1) R1 Interface Status

```
R1>show ip int brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES manual down                  down
GigabitEthernet0/1     192.168.1.1     YES manual up                    up
Vlan1                  unassigned      YES unset  administratively down down
```

2) S1 VLAN Config

```
S1>show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/1, Fa0/2, Fa0/3, Fa0/4
                                                Fa0/5, Fa0/6, Fa0/7, Fa0/8
                                                Fa0/9, Fa0/10, Fa0/11, Fa0/12
                                                Fa0/13, Fa0/14, Fa0/15, Fa0/16
                                                Fa0/17, Fa0/18, Fa0/19, Fa0/20
                                                Fa0/21, Fa0/22, Fa0/23, Fa0/24
                                                Gig0/1, Gig0/2
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
```

3) PC1 Connectivity

```
C:\>ping 192.168.1.11

Pinging 192.168.1.11 with 32 bytes of data:

Reply from 192.168.1.11: bytes=32 time=46ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time<1ms TTL=128
Reply from 192.168.1.11: bytes=32 time=7ms TTL=128

Ping statistics for 192.168.1.11:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 46ms, Average = 13ms
```

4) ARP Resolution

```
C:\>arp -a
  Internet Address      Physical Address      Type
  192.168.1.1           0004.9aec.de02        dynamic
  192.168.1.2           0060.3e86.7626        dynamic
  192.168.1.11          0007.ec78.42cc        dynamic
```

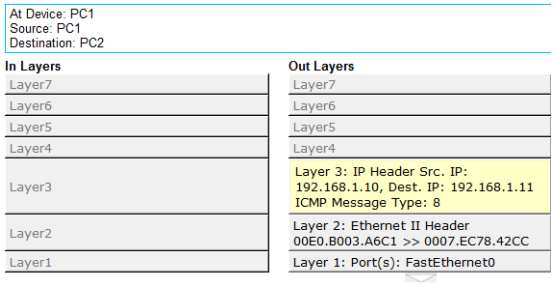
5) MAC Learning

```
S1#show mac address-table
          Mac Address Table
-------------------------------------------

Vlan    Mac Address       Type        Ports
----    -----------       --------    -----

   1    0004.9aec.de02    DYNAMIC     Gig0/1
   1    0007.ec78.42cc    DYNAMIC     Fa0/2
   1    00e0.b003.a6c1    DYNAMIC     Fa0/1
```

Checkpoint	Command Used	Expected Result	Actual Result	Pass/Fail
R1 Interface Status	show ip int brief	G0/1 UP/UP	G0/1 UP/UP	Pass
S1 VLAN Config	show vlan brief	VLAN 1 Active	VLAN 1 Active	Pass
PC1 Connectivity	ping 192.168.1.11	Success (4 replies)	Success	Pass
ARP Resolution	arp -a on PC1	2+ entries	3 entries shown	Pass
MAC Learning	show mac address-table on S1	2 dynamic entries	3 dynamic entries shown	Pass

Worksheet 2: Protocol Analysis

Capture this data during simulation:



1. ARP Request Frame:
   - Source MAC: 00E0.B003.A6C1
   - Destination MAC: FF:FF:FF:FF:FF:FF
   - ARP Opcode: 1 (Request)
2. ICMP Echo Packet:
   - Time to Live (TTL): _____128_____
   - Protocol Field: _____ICMP____ (Value = 1)
3. Encapsulation Chain: Data ⟶ ICMP Header ⟶ IP Header ⟶ Ethernet Header ⟶ Bits

Worksheet 3: Troubleshooting Journal

Problem Injected	Symptoms Observed	Diagnostic Commands Used	Root Cause	Solution
Wrong Subnet Mask	Ping to a remote network failed	ping, arp -a	Incorrect subnet mask	Restore correct subnet mask to 255.255.255.0
Interface Shutdown	PC unable to reach default gateway	show ip int brief	Router interface was shutdown	Enable interface using no shutdown
Incorrect Gateway	Unable to reach external networks	ipconfig, tracert	Default gateway pointed to a non-existent router	Restore correct default gateway to 192.168.1.1

🎓 ASSESSMENT QUESTIONS

1. When PC1 pings PC2, which protocol resolves the MAC address before ICMP operates?

   Ans ARP (Address Resolution Protocol) ที่ทำหน้าที่แปลง IP Address ปลายทางให้เป็น MAC Address เพื่อใช้สร้าง Frame ใน Layer ที่ 2

2. What would happen if you changed the subnet mask on PC1 to 255.255.255.128?

Ans ยังทำงานได้ตามปกติเนื่องจาก IP ของ PC1 และ PC2 ยังอยู่ภายใน Subnet Range เดียวกัน แต่ถ้ามีกรณีเครื่องที่มี IP มากกว่า .126 จะไม่สามารถสื่อสารด้วยกันได้

3. Why does the switch not need an IP address for basic frame forwarding?

Ans Switch เป็นอุปกรณ์ที่ทำงานใน Layer ที่ 2 ซึ่งใช้ MAC Address ในการตัดสินใจส่งข้อมูล ไม่ได้ใช้ IP Address ยกเว้นกรณีที่ต้องการบรหารจัดการ Switch จึงจะต้องตั้งค่า Management IP

4. Which OSI layers are involved when R1 forwards a packet between interfaces?

Ans มี 3 Layers หลัก

- Physical Layer รับส่งสัญญาณไฟฟ้าผ่านสาย
- Data Link Layer ถอด Frame Header เดิมออกไป และสร้าง Frame Header ใหม่
- Network Layer ตรวจสอบ IP ปลายทางและค้นหาเส้นทางใน Routing Table

5. What is the purpose of the TTL field in the IP header you observed?

Ans ป้องกันการเกิด Packet Loop เพราะ TTL จะลดค่าลงทุกครั้งที่ผ่าน Router เมื่อกลายเป็น 0 Packet จะถูกลบออกทันที